

## 6.3 Orthogonal projection

### 6.3.1 Nearest points

#### Theorem 6.3.1 — Projection theorem

Let  $\mathcal{S}$  be a subspace of  $\mathbb{R}^n$  and  $\mathbf{y} \in \mathbb{R}^n$ . There is exactly one  $\hat{\mathbf{y}} \in \mathcal{S}$  such that

$$\|\mathbf{y} - \hat{\mathbf{y}}\| \leq \|\mathbf{y} - \mathbf{u}\| \quad \text{for all } \mathbf{u} \in \mathcal{S}.$$

It is the unique  $\hat{\mathbf{y}} \in \mathcal{S}$  with  $\mathbf{y} - \hat{\mathbf{y}} \perp \mathcal{S}$ . Moreover, for every  $\mathbf{u} \in \mathcal{S}$ ,

$$\|\mathbf{y} - \mathbf{u}\|^2 = \|\mathbf{y} - \hat{\mathbf{y}}\|^2 + \|\hat{\mathbf{y}} - \mathbf{u}\|^2. \quad (6.3.1)$$

*Proof.* By Theorem 6.2.3 there is a unique decomposition  $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{v}$  with  $\hat{\mathbf{y}} \in \mathcal{S}$  and  $\mathbf{v} \in \mathcal{S}^\perp$ . For  $\mathbf{u} \in \mathcal{S}$ ,  $\mathbf{y} - \mathbf{u} = \mathbf{v} + (\hat{\mathbf{y}} - \mathbf{u})$ , where  $\mathbf{v} \perp \mathcal{S}$  and  $\hat{\mathbf{y}} - \mathbf{u} \in \mathcal{S}$ . The Pythagorean identity Eq. 6.2.3 then gives Eq. 6.3.1. The right-hand side is at least  $\|\mathbf{y} - \hat{\mathbf{y}}\|^2$ , with equality iff  $\mathbf{u} = \hat{\mathbf{y}}$ . So  $\hat{\mathbf{y}}$  is the unique nearest point. If some other  $\mathbf{w} \in \mathcal{S}$  also had  $\mathbf{y} - \mathbf{w} \perp \mathcal{S}$ , then  $\mathbf{y} = \mathbf{w} + (\mathbf{y} - \mathbf{w})$  would be a second decomposition, contradicting uniqueness. ■

The theorem has two halves, and it helps to keep them apart. The *variational* half says  $\hat{\mathbf{y}}$  minimizes a distance. The *orthogonality* half says the error  $\mathbf{y} - \hat{\mathbf{y}}$  is perpendicular to the subspace. The first half explains why least squares is a sensible estimation principle. The second is what we compute with, because orthogonality is a set of linear equations.

### 6.3.2 The projection as a matrix

Write  $P_{\mathcal{S}}\mathbf{y}$  for the nearest point  $\hat{\mathbf{y}}$ . The map  $\mathbf{y} \mapsto P_{\mathcal{S}}\mathbf{y}$  is linear: if  $\mathbf{y}_k = \hat{\mathbf{y}}_k + \mathbf{v}_k$  are the decompositions of  $\mathbf{y}_1$  and  $\mathbf{y}_2$ , then  $a\mathbf{y}_1 + b\mathbf{y}_2 = (a\hat{\mathbf{y}}_1 + b\hat{\mathbf{y}}_2) + (a\mathbf{v}_1 + b\mathbf{v}_2)$  is a decomposition of the same kind, and by uniqueness it is *the* decomposition. A linear map on  $\mathbb{R}^n$  is given by a unique  $n \times n$  matrix, since two matrices that agree on every vector agree on the standard basis vectors and are therefore equal.

#### Definition 6.3.1 — Orthogonal projection matrix

The **orthogonal projection** onto a subspace  $\mathcal{S}$  is the unique  $n \times n$  matrix  $P_{\mathcal{S}}$  such that, for every  $\mathbf{y}$ ,  $P_{\mathcal{S}}\mathbf{y} \in \mathcal{S}$  and  $\mathbf{y} - P_{\mathcal{S}}\mathbf{y} \perp \mathcal{S}$ . When  $\mathcal{S} = \mathbb{C}(\mathbf{X})$  for the model matrix of a linear model, we write  $\mathbf{M} = P_{\mathbb{C}(\mathbf{X})}$ .

The most useful formula for  $P_{\mathcal{S}}$  uses an orthonormal basis.

### Proposition 6.3.2

Let the columns of the  $n \times r$  matrix  $\mathbb{Q}$  be an orthonormal basis for  $\mathcal{S}$ , so that  $\mathbb{Q}\mathbb{T}\mathbb{Q} = \mathbf{I}_r$ . Then

$$\mathbf{P}_S = \mathbb{Q}\mathbb{Q}\mathbf{T} = \sum_{i=1}^r \mathbf{u}_i \mathbf{u}_i \mathbf{T},$$

where  $\mathbf{u}_1, \dots, \mathbf{u}_r$  are the columns of  $\mathbb{Q}$ .

*Proof.* For any  $\mathbf{y}$ ,  $\mathbb{Q}\mathbb{Q}\mathbf{T}\mathbf{y} = \mathbb{Q}(\mathbb{Q}\mathbf{T}\mathbf{y}) \in \mathbb{C}(\mathbb{Q}) = \mathcal{S}$ , and  $\mathbb{Q}\mathbf{T}(\mathbf{y} - \mathbb{Q}\mathbb{Q}\mathbf{T}\mathbf{y}) = \mathbb{Q}\mathbf{T}\mathbf{y} - \mathbb{Q}\mathbf{T}\mathbf{y} = \mathbf{0}$ , so the difference is orthogonal to every column of  $\mathbb{Q}$  and hence to  $\mathcal{S}$ . By uniqueness,  $\mathbb{Q}\mathbb{Q}\mathbf{T} = \mathbf{P}_S$ . ■

#### Example. (Projections onto a line)

For  $\mathcal{S} = \text{span}(\mathbf{v})$  with  $\mathbf{v} \neq \mathbf{0}$ , the single vector  $\mathbf{v}/\|\mathbf{v}\|$  is an orthonormal basis, so

$$\mathbf{P}_S = \frac{\mathbf{v}\mathbf{v}\mathbf{T}}{\mathbf{v}\mathbf{T}\mathbf{v}}, \quad \mathbf{P}_S \mathbf{y} = \frac{\mathbf{v}\mathbf{T}\mathbf{y}}{\mathbf{v}\mathbf{T}\mathbf{v}} \mathbf{v}.$$

With  $\mathbf{v} = \mathbf{1}$  this gives  $\mathbf{P} = n^{-1}\mathbf{1}\mathbf{1}\mathbf{T}$ , the matrix with every entry  $1/n$ , and  $\mathbf{P}\mathbf{y} = \bar{y}\mathbf{1}$ . *The sample mean is a projection:* fitting the intercept-only model replaces every observation by  $\bar{y}$ , and the residuals  $y_i - \bar{y}$  form a vector in  $\mathbf{1}^\perp$ , the set of vectors whose entries sum to zero.

## 6.3.3 Characterizing projection matrices

Which matrices are orthogonal projections? [Proposition 6.3.2](#) shows that every projection has the form  $\mathbb{Q}\mathbb{Q}\mathbf{T}$ , so it is *symmetric*, and  $\mathbb{Q}\mathbb{Q}\mathbf{T}\mathbb{Q}\mathbb{Q}\mathbf{T} = \mathbb{Q}\mathbf{I}\mathbb{Q}\mathbf{T} = \mathbb{Q}\mathbb{Q}\mathbf{T}$ , so it is *idempotent*. The converse is the basic recognition theorem.

### Theorem 6.3.3

An  $n \times n$  matrix  $\mathbf{P}$  is the orthogonal projection onto some subspace iff  $\mathbf{P}\mathbf{T} = \mathbf{P}$  and  $\mathbf{P}^2 = \mathbf{P}$ . In that case the subspace is  $\mathbb{C}(\mathbf{P})$ , and  $\mathbf{I} - \mathbf{P}$  is the orthogonal projection onto  $\mathbb{C}(\mathbf{P})^\perp$ .

*Proof.* Necessity was shown above. For sufficiency, let  $\mathbf{P}$  be symmetric and idempotent and let  $\mathbf{y} \in \mathbb{R}^n$ . Certainly  $\mathbf{P}\mathbf{y} \in \mathbb{C}(\mathbf{P})$ . For any  $\mathbf{u} = \mathbf{P}\mathbf{b} \in \mathbb{C}(\mathbf{P})$ ,

$$\mathbf{u}\mathbf{T}(\mathbf{y} - \mathbf{P}\mathbf{y}) = \mathbf{b}\mathbf{T}\mathbf{P}\mathbf{T}(\mathbf{I} - \mathbf{P})\mathbf{y} = \mathbf{b}\mathbf{T}(\mathbf{P} - \mathbf{P}^2)\mathbf{y} = 0,$$

so  $\mathbf{y} - \mathbf{P}\mathbf{y} \perp \mathbb{C}(\mathbf{P})$  and  $\mathbf{P} = \mathbf{P}_{\mathbb{C}(\mathbf{P})}$ . For the last claim,  $\mathbf{I} - \mathbf{P}$  is symmetric and idempotent, since  $(\mathbf{I} - \mathbf{P})^2 = \mathbf{I} - 2\mathbf{P} + \mathbf{P}^2 = \mathbf{I} - \mathbf{P}$ . It sends  $\mathbf{y}$  to  $\mathbf{y} - \mathbf{P}\mathbf{y} \in \mathbb{C}(\mathbf{P})^\perp$ , and  $\mathbf{y} - (\mathbf{I} - \mathbf{P})\mathbf{y} = \mathbf{P}\mathbf{y} \perp \mathbb{C}(\mathbf{P})^\perp$ .

Before this theorem, checking that a matrix computes a projection meant thinking about subspaces. Afterwards it is two matrix identities. Also, a projection is determined by its range: if  $\mathbf{P}_1, \mathbf{P}_2$  are symmetric idempotent with  $\mathbb{C}(\mathbf{P}_1) = \mathbb{C}(\mathbf{P}_2)$ , then  $\mathbf{P}_1 = \mathbf{P}_2$ , because both are the projection onto the same subspace. ■

### Proposition 6.3.4 — Spectrum, rank and trace

Let  $\mathbf{P}$  be the orthogonal projection onto an  $r$ -dimensional subspace  $\mathcal{S}$ . Then:

- (a) every eigenvalue of  $\mathbf{P}$  is 0 or 1, and 1 has multiplicity  $r$ ;
- (b)  $\text{rank}(\mathbf{P}) = \text{tr}(\mathbf{P}) = r$ ;
- (c)  $\mathbf{P}\mathbf{v} = \mathbf{v}$  iff  $\mathbf{v} \in \mathcal{S}$ , and  $\mathbf{P}\mathbf{v} = \mathbf{0}$  iff  $\mathbf{v} \in \mathcal{S}^\perp$ ;
- (d)  $\mathbf{v}^T \mathbf{P} \mathbf{v} = \|\mathbf{P}\mathbf{v}\|^2 \leq \|\mathbf{v}\|^2$ , so  $\mathbf{P}$  and  $\mathbf{I} - \mathbf{P}$  are nonnegative definite.

*Proof.* Write  $\mathbf{P} = \mathbf{Q}\mathbf{Q}^T$  as in Proposition 6.3.2, and extend  $\mathbf{Q}$  to an orthogonal matrix  $\mathbf{U} = [\mathbf{Q}, \mathbf{Q}_\perp]$ . Then  $\mathbf{P} = \mathbf{U} \text{diag}(\mathbf{I}_r, \mathbf{0}) \mathbf{U}^T$  is a spectral decomposition, which gives (a). Rank and trace of a symmetric matrix are the number of nonzero eigenvalues and their sum, which gives (b). Directly,  $\text{tr}(\mathbf{Q}\mathbf{Q}^T) = \text{tr}(\mathbf{Q}^T\mathbf{Q}) = \text{tr}(\mathbf{I}_r) = r$ . Part (c) restates the projection theorem: the nearest point of  $\mathcal{S}$  to a vector already in  $\mathcal{S}$  is the vector itself, and  $\mathbf{v} \perp \mathcal{S}$  iff  $\mathbf{0}$  is its nearest point. For (d),  $\mathbf{v}^T \mathbf{P} \mathbf{v} = \mathbf{v}^T \mathbf{P}^T \mathbf{P} \mathbf{v}$ , and  $\|\mathbf{v}\|^2 = \|\mathbf{P}\mathbf{v}\|^2 + \|(\mathbf{I} - \mathbf{P})\mathbf{v}\|^2$  by Pythagoras. ■

The identity  $\text{rank} = \text{tr}$  gets a lot of use. The dimension of a subspace, which is the “degrees of freedom” of every sum of squares in this book, can be read off as a sum of diagonal entries without finding a basis.

## 6.3.4 Generalized inverses

Proposition 6.3.2 needs an orthonormal basis. In statistics the subspace usually comes as the column space of a model matrix  $\mathbf{X}$  whose columns are not orthonormal and may not even be independent. To write  $\mathbf{M}$  directly in terms of  $\mathbf{X}$  we need a substitute for  $(\mathbf{X}^T\mathbf{X})^{-1}$  that exists even when  $\mathbf{X}^T\mathbf{X}$  is singular.

### Definition 6.3.2 — Generalized inverse

A **generalized inverse** of an  $m \times k$  matrix  $\mathbf{A}$  is any  $k \times m$  matrix  $\mathbf{A}^-$  such that  $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$ .

If  $\mathbf{A}$  is square and nonsingular, multiplying  $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$  on both sides by  $\mathbf{A}^{-1}$  gives  $\mathbf{A}^- = \mathbf{A}^{-1}$ , so the generalized inverse is unique and equal to the inverse. When  $\mathbf{A}$  is singular there are infinitely many.

### Proposition 6.3.5 — Existence

Every matrix has a generalized inverse. If  $\mathbf{A}$  has rank  $r$  and is permuted so that

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}$$

with  $\mathbf{A}_{11}$  an  $r \times r$  nonsingular block, then  $\mathbf{A}^- = \begin{pmatrix} \mathbf{A}_{11}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$  (permuted back) is a generalized inverse.

*Proof.* A rank- $r$  matrix has  $r$  independent rows and  $r$  independent columns, and the submatrix where they cross is nonsingular (Chapter 1). This guarantees the block form. Since the first  $r$  columns are a basis for the column space,  $\begin{pmatrix} \mathbf{A}_{12} \\ \mathbf{A}_{22} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11} \\ \mathbf{A}_{21} \end{pmatrix} \mathbf{B}$  for some  $\mathbf{B}$ , so  $\mathbf{B} = \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$  and  $\mathbf{A}_{22} = \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$ . Multiplying out,

$$\mathbf{A} \mathbf{A}^- \mathbf{A} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & \mathbf{0} \end{pmatrix} \mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \end{pmatrix} = \mathbf{A}.$$

■

Among the many generalized inverses, one is singled out by extra symmetry conditions. The **Moore–Penrose inverse**  $\mathbf{A}^+$  is the unique matrix satisfying  $\mathbf{A} \mathbf{A}^+ \mathbf{A} = \mathbf{A}$ ,  $\mathbf{A}^+ \mathbf{A} \mathbf{A}^+ = \mathbf{A}^+$ , and both  $\mathbf{A} \mathbf{A}^+$  and  $\mathbf{A}^+ \mathbf{A}$  symmetric (Penrose 1955). Section 6.10 computes it from the singular value decomposition. For projections we will need none of these extra conditions, and that is the point of the next result.

### Lemma 6.3.6

Let  $\mathbf{X}$  be  $n \times p$  and let  $\mathbf{G}$  be any generalized inverse of  $\mathbf{X} \mathbf{T} \mathbf{X}$ . Then:

- (a)  $\mathbf{X} \mathbf{G} \mathbf{X} \mathbf{T} \mathbf{X} = \mathbf{X}$  and  $\mathbf{X} \mathbf{T} \mathbf{X} \mathbf{G} \mathbf{X} \mathbf{T} = \mathbf{X} \mathbf{T}$ ;
- (b)  $\mathbf{X} \mathbf{G} \mathbf{X} \mathbf{T}$  does not depend on the choice of generalized inverse;
- (c)  $\mathbf{X} \mathbf{G} \mathbf{X} \mathbf{T}$  is symmetric, even when  $\mathbf{G}$  is not.

*Proof.* By Corollary 6.2.4(b),  $\mathbb{C}(\mathbf{X} \mathbf{T}) = \mathbb{C}(\mathbf{X} \mathbf{T} \mathbf{X})$ , so there is a  $p \times n$  matrix  $\mathbf{B}$  with  $\mathbf{X} \mathbf{T} = \mathbf{X} \mathbf{T} \mathbf{X} \mathbf{B}$ . Transposing,  $\mathbf{X} = \mathbf{B} \mathbf{T} \mathbf{X} \mathbf{T} \mathbf{X}$ . Then

$$\mathbf{X} \mathbf{G} \mathbf{X} \mathbf{T} \mathbf{X} = \mathbf{B} \mathbf{T} (\mathbf{X} \mathbf{T} \mathbf{X}) \mathbf{G} (\mathbf{X} \mathbf{T} \mathbf{X}) = \mathbf{B} \mathbf{T} \mathbf{X} \mathbf{T} \mathbf{X} = \mathbf{X},$$

which is the first identity in (a). The second follows the same way, using  $\mathbf{X} \mathbf{T} = \mathbf{X} \mathbf{T} \mathbf{X} \mathbf{B}$  on the right. For (b) and (c),

$$\mathbf{X} \mathbf{G} \mathbf{X} \mathbf{T} = \mathbf{B} \mathbf{T} (\mathbf{X} \mathbf{T} \mathbf{X}) \mathbf{G} (\mathbf{X} \mathbf{T} \mathbf{X}) \mathbf{B} = \mathbf{B} \mathbf{T} \mathbf{X} \mathbf{T} \mathbf{X} \mathbf{B},$$

which involves no  $\mathbf{G}$  at all, and  $(\mathbf{B} \mathbf{T} \mathbf{X} \mathbf{T} \mathbf{X} \mathbf{B})^T = \mathbf{B} \mathbf{T} \mathbf{X} \mathbf{T} \mathbf{X} \mathbf{B}$ .

■

### Theorem 6.3.7

For any  $n \times p$  matrix  $\mathbf{X}$  and any generalized inverse  $(\mathbf{X}\mathbf{T}\mathbf{X})^-$ ,

$$\mathbf{M} = \mathbf{X}(\mathbf{X}\mathbf{T}\mathbf{X})^- \mathbf{X}\mathbf{T}$$

is the orthogonal projection onto  $\mathbb{C}(\mathbf{X})$ . In particular  $\text{rank}(\mathbf{M}) = \text{tr}(\mathbf{M}) = \text{rank}(\mathbf{X})$ . If  $\mathbf{X}$  has full column rank,  $\mathbf{M} = \mathbf{X}(\mathbf{X}\mathbf{T}\mathbf{X})^{-1} \mathbf{X}\mathbf{T}$ .

*Proof.* By Lemma 6.3.6,  $\mathbf{M}$  is symmetric and  $\mathbf{M}^2 = \mathbf{X}(\mathbf{X}\mathbf{T}\mathbf{X})^- [\mathbf{X}\mathbf{T}\mathbf{X}(\mathbf{X}\mathbf{T}\mathbf{X})^- \mathbf{X}\mathbf{T}] = \mathbf{X}(\mathbf{X}\mathbf{T}\mathbf{X})^- \mathbf{X}\mathbf{T} = \mathbf{M}$ . By Theorem 6.3.3,  $\mathbf{M}$  is the projection onto  $\mathbb{C}(\mathbf{M})$ . Clearly  $\mathbb{C}(\mathbf{M}) \subseteq \mathbb{C}(\mathbf{X})$ , and  $\mathbf{M}\mathbf{X} = \mathbf{X}$  (again by the lemma) shows  $\mathbb{C}(\mathbf{X}) \subseteq \mathbb{C}(\mathbf{M})$ . The rank statement is Proposition 6.3.4(b). ■

It is worth pausing on how strong Theorem 6.3.7 is. The matrix  $(\mathbf{X}\mathbf{T}\mathbf{X})^-$  can be almost anything: symmetric or not, reflexive or not, with entries of any size. The sandwich  $\mathbf{X}(\cdot)\mathbf{X}\mathbf{T}$  removes all of that freedom. We will see in Section 6.4 that this is exactly the difference between coefficients and fitted values, made concrete.

#### Example. (The one-way layout)

In the setting of Example (A rank-deficient layout), sort the observations by group and let group  $k$  have  $n_k$  members. The model space is spanned by the orthogonal vectors  $\mathbf{z}_1, \dots, \mathbf{z}_g$ , so by Proposition 6.3.2 with  $\mathbf{u}_k = \mathbf{z}_k / \sqrt{n_k}$ ,

$$\mathbf{M} = \sum_{k=1}^g \frac{\mathbf{z}_k \mathbf{z}_k^T}{n_k} = \begin{pmatrix} n_1^{-1} \mathbf{1}_{n_1} \mathbf{1}_{n_1}^T & & \\ & \ddots & \\ & & n_g^{-1} \mathbf{1}_{n_g} \mathbf{1}_{n_g}^T \end{pmatrix},$$

a block-diagonal matrix of within-group averaging operators.  $\mathbf{M}\mathbf{y}$  replaces each observation by its group mean. The intercept column adds nothing to  $\mathbb{C}(\mathbf{X})$ , so it does not appear, and  $\text{tr}(\mathbf{M}) = \sum_k n_k/n_k = g = \text{rank}(\mathbf{X})$ . We got here without a generalized inverse. Theorem 6.3.7 guarantees that any generalized inverse of the singular  $(g+1) \times (g+1)$  matrix  $\mathbf{X}\mathbf{T}\mathbf{X}$  produces this same matrix. Section 6.4 checks this numerically.

## 6.3.5 Sums and products of projections

Complicated subspaces are often built from simpler ones, and their projections are built the same way.

### Theorem 6.3.8

Let  $\mathbf{P}_1, \mathbf{P}_2$  be orthogonal projections. Then  $\mathbf{P}_1 + \mathbf{P}_2$  is an orthogonal projection iff  $\mathbf{P}_1 \mathbf{P}_2 = \mathbf{0}$ , that is, iff  $\mathbb{C}(\mathbf{P}_1) \perp \mathbb{C}(\mathbf{P}_2)$ . In that case it projects onto  $\mathbb{C}(\mathbf{P}_1) \oplus \mathbb{C}(\mathbf{P}_2)$ .

*Proof.*  $P_1 + P_2$  is always symmetric, and  $(P_1 + P_2)^2 = P_1 + P_2 + P_1P_2 + P_2P_1$ . So it is idempotent iff  $P_1P_2 + P_2P_1 = 0$ . If  $P_1P_2 = 0$ , then transposing gives  $P_2P_1 = 0$  and the condition holds. Conversely, suppose  $P_1P_2 + P_2P_1 = 0$ . Multiplying on the left by  $P_1$  gives  $P_1P_2 + P_1P_2P_1 = 0$ , and on the right by  $P_1$  gives  $P_1P_2P_1 + P_2P_1 = 0$ . Subtracting,  $P_1P_2 = P_2P_1$ , and then  $2P_1P_2 = 0$ . The equivalence with orthogonality of ranges holds because  $P_1P_2 = 0$  says exactly that every vector of  $\mathbb{C}(P_2)$  is sent to  $0$  by  $P_1$ , that is (Proposition 6.3.4(c)), lies in  $\mathbb{C}(P_1)^\perp$ . Finally,  $\mathbb{C}(P_1 + P_2) \subseteq \mathbb{C}(P_1) + \mathbb{C}(P_2)$ , and for  $u = P_1u + P_2u$  in that sum we have  $(P_1 + P_2)u = u$ , so the ranges agree. ■

By induction, if  $P_1, \dots, P_k$  are projections with  $P_iP_j = 0$  for  $i \neq j$ , their sum projects onto the orthogonal direct sum of their ranges. When the sum is  $\mathbf{I}$ , we get an orthogonal decomposition

$$y = P_1y + \dots + P_ky, \quad \|y\|^2 = \|P_1y\|^2 + \dots + \|P_ky\|^2.$$

Every analysis of variance table in this book is a table of such a decomposition.

Products behave differently.  $P_1P_2$  need not be symmetric or idempotent, and it is generally not a projection. Exercise (B1) shows that it is one iff  $P_1$  and  $P_2$  commute, in which case it projects onto  $\mathbb{C}(P_1) \cap \mathbb{C}(P_2)$ . The most important instance is the nested case  $\mathbb{C}(P_2) \subseteq \mathbb{C}(P_1)$ , treated in Section 6.5.

## 6.3.6 Exercises

### 6.3.6.1 A. Check your understanding

#### Exercise A1.

Let  $\mathcal{S} = \text{span}\{(1, 1, 0)\mathbf{T}, (0, 1, 1)\mathbf{T}\}$ . Compute  $P_{\mathcal{S}}$  in two ways: by Gram–Schmidt and Proposition 6.3.2, and by Theorem 6.3.7. Verify symmetry, idempotence and  $\text{tr } P_{\mathcal{S}} = 2$ . Find  $\mathcal{S}^\perp$  and check that  $\mathbf{I} - P_{\mathcal{S}}$  projects onto it.

#### Exercise A2.

Show that a symmetric matrix whose eigenvalues all lie in  $\{0, 1\}$  is an orthogonal projection. Give a non-symmetric  $2 \times 2$  matrix with eigenvalues 0 and 1 that is not idempotent, and one that is idempotent.

#### Exercise A3.

Let  $P$  be an orthogonal projection and  $a, b$  real numbers. Show that the eigenvalues of  $aP + b\mathbf{I}$  are  $b$  and  $a + b$ . Show that when  $b \neq 0$  and  $a + b \neq 0$ ,

$$(aP + b\mathbf{I})^{-1} = \frac{1}{b} \left( \mathbf{I} - \frac{a}{a+b} P \right).$$

Use this to invert the equicorrelation matrix  $(1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}\mathbf{T}$ .

**Exercise A4.**

Show that if  $\mathbf{G}$  is a generalized inverse of the symmetric matrix  $\mathbf{X}\mathbf{T}\mathbf{X}$ , then so is  $\mathbf{G}\mathbf{T}$ , and so is  $\mathbf{G}\mathbf{X}\mathbf{T}\mathbf{X}\mathbf{G}\mathbf{T}$ . Which of the four Penrose conditions does the last one satisfy?

**6.3.6.2 B. Practice****Exercise B1.**

Let  $\mathbf{P}_1, \mathbf{P}_2$  be orthogonal projections. Show that  $\mathbf{P}_1\mathbf{P}_2$  is an orthogonal projection iff  $\mathbf{P}_1\mathbf{P}_2 = \mathbf{P}_2\mathbf{P}_1$ , and that in this case it projects onto  $\mathbb{C}(\mathbf{P}_1) \cap \mathbb{C}(\mathbf{P}_2)$ . *Hint:* for the range, show  $\mathbf{P}_1\mathbf{P}_2\mathbf{u} = \mathbf{u}$  iff  $\mathbf{u}$  lies in both subspaces.

*Solution.* If  $\mathbf{P}_1\mathbf{P}_2 = \mathbf{P}_2\mathbf{P}_1$ , then  $(\mathbf{P}_1\mathbf{P}_2)\mathbf{T} = \mathbf{P}_2\mathbf{P}_1 = \mathbf{P}_1\mathbf{P}_2$ , and  $(\mathbf{P}_1\mathbf{P}_2)^2 = \mathbf{P}_1\mathbf{P}_2\mathbf{P}_2\mathbf{P}_1 = \mathbf{P}_1\mathbf{P}_2\mathbf{P}_1 = \mathbf{P}_1\mathbf{P}_1\mathbf{P}_2 = \mathbf{P}_1\mathbf{P}_2$ . So the product is symmetric and idempotent. Conversely, if  $\mathbf{P}_1\mathbf{P}_2$  is an orthogonal projection, it is symmetric, so  $\mathbf{P}_1\mathbf{P}_2 = (\mathbf{P}_1\mathbf{P}_2)\mathbf{T} = \mathbf{P}_2\mathbf{P}_1$ . For the range: if  $\mathbf{u}$  lies in both subspaces, then  $\mathbf{P}_1\mathbf{P}_2\mathbf{u} = \mathbf{P}_1\mathbf{u} = \mathbf{u}$ . If  $\mathbf{P}_1\mathbf{P}_2\mathbf{u} = \mathbf{u}$ , then  $\mathbf{u} \in \mathbb{C}(\mathbf{P}_1)$ , and also  $\mathbf{u} = \mathbf{P}_2\mathbf{P}_1\mathbf{u} \in \mathbb{C}(\mathbf{P}_2)$ . A projection's range is the set of vectors it fixes, so  $\mathbb{C}(\mathbf{P}_1\mathbf{P}_2) = \mathbb{C}(\mathbf{P}_1) \cap \mathbb{C}(\mathbf{P}_2)$ .

**Exercise B2.**

Let  $\mathbf{G}_0$  be a generalized inverse of  $\mathbf{A}$ .

- Show that  $\mathbf{G}_0 + (\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{U} + \mathbf{V}(\mathbf{I} - \mathbf{A}\mathbf{G}_0)$  is a generalized inverse of  $\mathbf{A}$  for all conformable  $\mathbf{U}, \mathbf{V}$ .
- Show that every generalized inverse  $\mathbf{G}$  of  $\mathbf{A}$  has this form. *Hint:* take  $\mathbf{U} = \mathbf{G} - \mathbf{G}_0$  and  $\mathbf{V} = \mathbf{G}_0\mathbf{A}(\mathbf{G} - \mathbf{G}_0)$ , and use  $\mathbf{A}(\mathbf{G} - \mathbf{G}_0)\mathbf{A} = \mathbf{0}$ .
- Conclude that a matrix has a unique generalized inverse iff it is square and non-singular.

*Solution. Parts a,b.* For (a), multiply out:  $\mathbf{A}[\mathbf{G}_0 + (\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{U} + \mathbf{V}(\mathbf{I} - \mathbf{A}\mathbf{G}_0)]\mathbf{A} = \mathbf{A}\mathbf{G}_0\mathbf{A} + (\mathbf{A} - \mathbf{A}\mathbf{G}_0\mathbf{A})\mathbf{U}\mathbf{A} + \mathbf{A}\mathbf{V}(\mathbf{A} - \mathbf{A}\mathbf{G}_0\mathbf{A}) = \mathbf{A}$ . For (b), put  $\mathbf{\Delta} = \mathbf{G} - \mathbf{G}_0$ , so that  $\mathbf{A}\mathbf{\Delta}\mathbf{A} = \mathbf{A} - \mathbf{A} = \mathbf{0}$ . With  $\mathbf{U} = \mathbf{\Delta}$  and  $\mathbf{V} = \mathbf{G}_0\mathbf{A}\mathbf{\Delta}$ ,

$$(\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{\Delta} + \mathbf{G}_0\mathbf{A}\mathbf{\Delta}(\mathbf{I} - \mathbf{A}\mathbf{G}_0) = \mathbf{\Delta} - \mathbf{G}_0\mathbf{A}\mathbf{\Delta} + \mathbf{G}_0\mathbf{A}\mathbf{\Delta} - \mathbf{G}_0(\mathbf{A}\mathbf{\Delta}\mathbf{A})\mathbf{G}_0 = \mathbf{\Delta},$$

so the formula reproduces  $\mathbf{G}_0 + \mathbf{\Delta} = \mathbf{G}$ .

**Exercise B3.**

Using the singular value decomposition, show that  $(\mathbf{X}\mathbf{T}\mathbf{X})^+\mathbf{X}\mathbf{T} = \mathbf{X}^+$  and  $\mathbf{X}\mathbf{X}^+ = \mathbf{M}$ . Show also that  $\mathbf{X}^+ = (\mathbf{X}\mathbf{T}\mathbf{X})^{-1}\mathbf{X}\mathbf{T}$  when  $\mathbf{X}$  has full column rank.

**6.3.6.3 C. Going deeper**

**Exercise C1.**

Show that  $\mathbf{M}$  is the only  $n \times n$  matrix  $\mathbf{P}$  satisfying  $\mathbf{P}\mathbf{X} = \mathbf{X}$ ,  $\mathbf{P}\mathbf{T} = \mathbf{P}$  and  $\text{rank}(\mathbf{P}) = \text{rank}(\mathbf{X})$ . Is the rank condition necessary? Is symmetry?